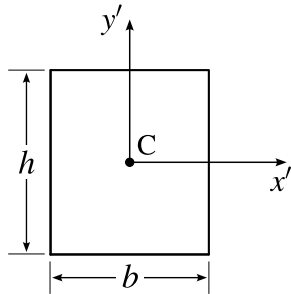


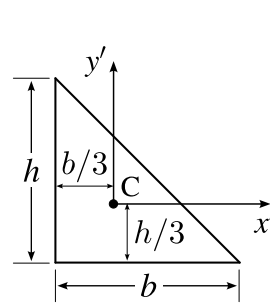
1) Centróide, momentos e produto de inércia de área

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA} \quad \bar{y} = \frac{\int \tilde{y} dA}{\int dA} \quad \bar{z} = \frac{\int \tilde{z} dA}{\int dA}$$

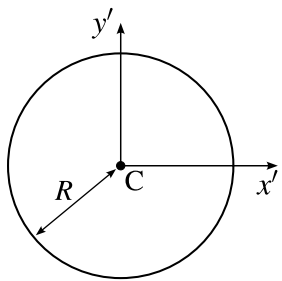
$$I_x = \int \tilde{y}^2 dA \quad I_y = \int \tilde{x}^2 dA \quad I_{xy} = \int \tilde{x} \tilde{y} dA \quad J_O = I_x + I_y \quad k = \sqrt{\frac{I}{A}}$$



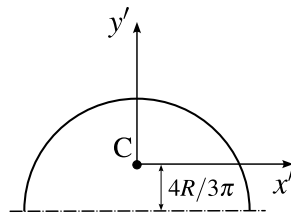
$$\begin{cases} I_{x'} = \frac{bh^3}{12} \\ I_{y'} = \frac{hb^3}{12} \\ I_{x'y'} = 0 \end{cases}$$



$$\begin{cases} I_{x'} = \frac{bh^3}{36} \\ I_{y'} = \frac{hb^3}{36} \\ I_{x'y'} = -\frac{b^2h^2}{72} \end{cases}$$



$$\begin{cases} I_{x'} = \frac{\pi R^4}{4} \\ I_{y'} = \frac{\pi R^4}{4} \\ I_{x'y'} = 0 \end{cases}$$



$$\begin{cases} I_{x'} = \frac{\pi R^4}{8} - \frac{8R^4}{9\pi} \\ I_{y'} = \frac{\pi R^4}{8} \\ I_{x'y'} = 0 \end{cases}$$

2) Teorema dos eixos paralelos

$$I_x = I_{x'} + A\bar{y}^2$$

$$I_y = I_{y'} + A\bar{x}^2$$

$$I_{xy} = I_{x'y'} + A\bar{x}\bar{y}$$

4) Teorema de Pappus-Guldinus

$$S = \theta \bar{r} L \quad V = \theta \bar{r} A$$

5) Carregamentos

$$F_R = \int p(x, y) dA \quad \bar{x} = \frac{\int p(x, y) x dA}{F_R} \quad \bar{y} = \frac{\int p(x, y) y dA}{F_R}$$

$$w(x) = p(x)b \quad F_R = \int w(x) dx \quad \bar{x} = \frac{\int x w(x) dx}{F_R} \quad \bar{y} = \frac{\int y w(x) dx}{F_R}$$

3) Momentos e produto de inércia em relação a eixos inclinados e direções principais

$$I_{med} = \frac{I_x + I_y}{2} \quad I_{sdif} = \frac{I_x - I_y}{2}$$

$$I_u = I_{med} + I_{sdif} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_v = I_{med} - I_{sdif} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{uv} = I_{sdif} \sin 2\theta + I_{xy} \cos 2\theta$$

$$\tan 2\theta_p = -\frac{I_{xy}}{I_{sdif}}$$